



What's New in Dolby® Media Producer Suite v2.0

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Introduction to What's New in Dolby Media Producer Suite

This documentation provides information about the Dolby® Media Producer Suite v2.0 release, including installation instructions, improvements, and known issues.

1.1 About Dolby Media Producer v2.0

Dolby Media Producer Suite v2.0 includes updates for Dolby Media Encoder Client, Dolby Media Encoder Server, and Dolby Media Decoder.

In this release, Dolby Media Encoder adds the ability to encode from a Dolby Atmos® master file set (home theater master), to create a Dolby TrueHD or Dolby Digital Plus™ bitstream with Dolby Atmos, or a Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstream.

In this release, Dolby Media Decoder supports decoding, playback, and monitoring of Dolby TrueHD and Dolby Digital Plus bitstreams carrying Dolby Atmos audio and associated metadata, and standard channel-based Dolby TrueHD, Dolby Digital Plus, Dolby Digital, MLP (for DVD audio) bitstreams as well.



Note: As of Dolby Media Producer Suite v1.1.2, Dolby Media Encoder SE (stand-alone) version is no longer included, as the more-powerful Client/Server version is now installed in all configurations. For more information, see [Section 2.5](#).

1.2 Contacting Dolby

You can contact Dolby regarding this product and its supporting documentation.

If you have technical questions about this product, please contact dmpSupport@dolby.com.

If you have questions or comments about this document, please contact [technical publications](#).

Installing Dolby Media Producer Suite

Use the Dolby® Media Producer Suite installer package provided by Dolby to install Dolby Media Producer Suite v2.0 applications. Dolby Media Producer Suite v2.0 includes updates for Dolby Media Encoder Client, Dolby Media Encoder Server, and Dolby Media Decoder.

2.1 Installation Overview

Complete installation of Dolby Media Producer Suite v2.0 includes authorizing the Dolby Media Producer Suite v2.X license on your iLok™ (if it is not already authorized), downloading and installing the MATLAB® Compiler Runtime (MCR) application release R2014b (8.4), reviewing considerations for your current software, and installing the software.

Before starting installation, review [System Requirements](#).

Dolby Media Producer Suite v2.0 installs the following applications, which have been updated:

- Dolby Media Encoder Client v2.0
- Dolby Media Encoder Server v2.0
- Dolby Media Decoder v2.0



Note: Dolby Media Producer Suite v2.0 does not include updates for any other of the applications for older Dolby Media Producer Suite versions. See [Section 2.5](#).

2.2 System Requirements

Before installing Dolby Media Producer Suite, you should review the minimum system requirements and compatibility information, to verify that your system supports the release and that you are ready to begin installation.

Minimum System Requirements

Dolby Media Producer Suite has the following minimum system requirements:

- Computer: Mac® Apple® Pro 4,1 or more recent (6,1 preferred).
- Processor: 2 x 2.26 GHz Quad-Core Intel Xeon or better.
- RAM: Minimum 8 GB, 1066 MHz DDR3 or faster (16 GB DDR4 recommended for encoding to Dolby TrueHD with Dolby Atmos).
- Mac OS® version: Mac OS X® 10.8.5 or 10.9.5.

Dolby Media Producer v2.0 will run on Apple Mac OS X 10.10, but has not been fully tested for all legacy and Dolby Atmos® workflows.

- System HDD: 20 GB of free space per hour of source material.
- Data HDD: Greater than or equal to 256 GB free on additional drive for input and output media storage (SSD preferred).

Dolby Media Decoder Audio Hardware Compatibility

Dolby Media Decoder has been validated on the following audio hardware:

- Avid® Pro Tools|HDX 3
 - Pro Tools|HD MADI
 - Mac OS X 10.8.5
 - Pro Tools 10 drivers
 - Mac OS X 10.9.5
 - Pro Tools 11 drivers
- Avid Pro Tools|HD Native, Pro Tools|HD I/O, or Pro Tools|HD OMNI
 - Mac OS X 10.8.5
 - Pro Tools 10 drivers
 - Mac OS X 10.9.5
 - Pro Tools 11 drivers
- RME Fireface UFX
 - Firewire 400 and USB 2.0
 - Mac OS X 10.8.5 and 10.9.5
- MOTU® 16A
 - USB
 - Mac OS X 10.8.5 and 10.9.5
- Universal Audio Apollo
 - Firewire 800
 - Mac OS X 10.8.5

2.3 Authorizing the Dolby Media Producer Suite v2.X License on Your iLok

The Dolby Media Encoder Server and Dolby Media Decoder applications use PACE iLok antipiracy software. A Dolby Media Producer Suite v2.X license stored on an iLok USB key (not supplied) authorizes your system to use the server and decoder applications. Additionally, Dolby Media Encoder Client uses drivers included with PACE iLok License Manager.



Note: If you previously participated in our beta program for this software, a new iLok authorization for the official v2.0 release is required.

The Dolby sales team or your engineering support contact will provide you with the codes that are required to authorize all of the systems that you need to run in your location. You may install the Dolby Media Producer Suite v2.X on multiple computers, and move the iLok keys to other computers, as needed.

Before you activate your license, make sure that you meet all hardware and software prerequisites:

- Make sure you have a physical iLok USB smart key (not supplied) and that it is inserted on a USB port on the computer where Dolby Media Producer Suite will be installed.
- If you do not have an existing iLok account, go to ilok.com and follow the directions to sign up for a free account.
- If you do not have iLok License Manager v2.5.x or later installed on your computer, go to ilok.com and follow the directions to download and install the current version. The iLok License Manager, downloadable from the iLok website (ilok.com), enables you to manage the licenses for all of your iLok-protected applications in one place. The iLok License Manager also installs drivers required to run Dolby Media Producer Suite applications. The iLok License Manager must be installed before installing the Dolby Media Producer Suite.
- Make sure you have the codes that are required to authorize all of the systems that you need to run in your location. These were provided by the Dolby sales team or your engineering support contact.

To activate your licenses:

1. Launch the **iLok License Manager**.
2. In the iLok License Manager window, sign in to your account, and then click (highlight) your account name in the navigation bar.
3. Click the **Redeem License Code** navigation link at the top of the page (to the right of the search field), or select this option from the **Licenses** menu in the menu bar.
4. In the **Enter Activation Code** dialog, copy and paste one of the license codes that were provided by Dolby.
5. Click **Next** in the dialog.
6. In the **Select an Activation Location** dialog, review the name of the license activation to be redeemed, and then select the iLok you have inserted into the computer as the destination for the new license.
7. Click the **Activate** button in the lower right-hand corner.
8. When prompted to confirm the activation and the destination iLok, click **OK**.

9. Verify that the new Dolby Media Producer Suite v2.X license activation is shown in the list for your account, and then click the bar for your iLok in the left-hand column to show the list of licenses stored on that iLok.

You should see the Dolby Media Producer Suite v2.X license in that list, too.

10. Repeat the preceding steps with additional iLoks, using the remaining supplied coupon codes.



Note: If you intend to move the iLok keys to other computers in order to run Dolby Media Producer Suite, verify that the latest version of the iLok License Manager application is installed on those computers.

The iLok License Manager, downloadable from the iLok website (ilok.com), installs drivers required to run Dolby Media Producer Suite applications.

You are now ready to install the MATLAB Compiler Runtime application.

2.4 Installing the MATLAB Compiler Runtime (MCR) Application

Prior to installing and using Dolby Media Producer Suite v2.0, you must install the MATLAB Compiler Runtime (MCR) application from MathWorks®. The Dolby TrueHD encoder in Dolby Media Encoder v2.0 uses the MCR application to run parts of the encoding algorithm.



Note: Dolby Media Producer Suite v2.0 requires MATLAB Compiler Runtime (MCR) application release R2014b (8.4).

To install the MCR application:

1. Navigate to the MCR download website:
<http://www.mathworks.com/products/compiler/mcr/>
2. Download release R2014b (8.4) for Mac, Intel® 64 bit.
3. Double click the installer, and follow onscreen instructions.



Note: When prompted to choose a location for the application, choose the default location. If the MCR application is installed to a different location, it will not support the Dolby TrueHD encoder.

2.5 Considerations for Existing Dolby Media Producer Suite Software

If you are updating from a previous version of Dolby Media Producer Suite, we recommend that you review how updating affects existing Dolby Media Producer Suite software on your computer.

When you install Dolby Media Producer Suite v2.0, the existing versions of Dolby Media Encoder Client, Dolby Media Encoder Server, and Dolby Media Decoder are replaced.

Following installation, Dolby Media Encoder v2.0 is ready for encoding from a Dolby Atmos master file set or PCM source audio .wav or .aif files. Additionally, Dolby Media Decoder v2.0 is ready for decoding and monitoring Dolby TrueHD and Dolby Digital Plus™ bitstreams carrying Dolby Atmos audio and associated metadata, as well as Dolby TrueHD, Dolby Digital Plus, and Dolby Digital bitstreams.

If you have previously installed Dolby Media Producer Suite applications other than Dolby Media Encoder Client, Dolby Media Encoder Server, and Dolby Media Decoder, those applications will remain where they were installed. These include Dolby Datarate Estimator, Dolby Media Tools, Dolby Branchpoint Preview Player, and Dolby TrueHD Quality and Conformance Tool. You can continue to use these applications, although they have not been fully qualified to support Dolby Atmos content. Using Dolby Media Tools with Dolby TrueHD or Dolby Digital Plus bitstreams with Dolby Atmos is not supported.

Shortcuts and Quick-Launch Icons

If you have created any shortcuts to the default location of v1.1.x of Dolby Media Encoder or Dolby Media Decoder, or quick-launch icons on the Mac OS X dock, they will now work for the default location of the updated version. If you have created any shortcuts to the default location of v1.0.x of Dolby Media Encoder or Dolby Media Decoder, or quick-launch icons on the Mac OS X dock, they should be updated to point to the default location of the updated version (in /Applications/Dolby/Dolby Media Producer).

2.6 Installing Dolby Media Producer Suite v2.0

You can install Dolby Media Producer Suite v2.0 applications on any Apple Mac OS X 10.8 or 10.9 computer qualified and set up for Dolby Media Producer Suite.

For guidance on system setup after the software has been installed, refer to the user manual for each installed software application. You can access the user manual from within an application by choosing **Help** and then the name of the application manual.

To install Dolby Media Producer Suite on a Mac:

1. Make sure the downloaded compressed file (.zip) is on your local hard drive, and not on a network drive.
2. Double-click the downloaded .zip file to unzip it.
3. Double-click the Dolby Media Producer Suite installer package (.pkg) to begin installation.
4. Follow the onscreen instructions to install the software.
 - When the **Custom Install** dialog appears, click (check) the applications you would like to install.



Note: The Dolby Media Encoder Server is required to encode with the Dolby Media Encoder. It runs as a background process on the machine it is installed on.

Ensure that the machine with the Dolby Media Encoder Server installed has an iLok with the Dolby Media Producer Suite v2.X license loaded on it (see [Section 2.3](#)).

5. When prompted, restart your system to complete installation.

6. After installation has completed, verify that an iLok USB key with the licenses for Dolby Media Producer on it is inserted on a USB port on the computer (as described in [Section 2.3](#)).
The license authorizes your computer to use the Dolby Media Producer Suite software applications.
7. Repeat the preceding steps to install the suite on any other Mac OS X computers in your facility.
Ten license codes are provided so that you can run Dolby Media Encoder and Dolby Media Decoder on up to ten computers at one time. You may install the suite on more than ten different computers, and move the iLok keys to other computers, as needed.



Note: If you intend to move the iLok keys to other computers in order to run Dolby Media Producer Suite, verify that the iLok License Manager is installed on those computers.

You are now ready to use Dolby Media Producer Suite v2.0.

Dolby Media Encoder Features and Improvements

Dolby® Media Producer Suite v2.0 updates the Dolby Media Encoder software application to version 2.0.

3.1 Overview of Dolby Media Encoder New Features and Improvements

Dolby Media Encoder v2.0 includes new features and improvements, primarily focused on encoding from a Dolby Atmos® master file set (home theater master) or legacy (.wav) channel-based material.

New features and improvements with Dolby Media Encoder v2.0:

- Ability to encode from a Dolby Atmos master file set (home theater master) to create any of the following bitstreams:
 - Dolby TrueHD bitstream with Dolby Atmos
 - Dolby Digital Plus™ bitstream with Dolby Atmos
 - Dolby TrueHD 7.1 bitstream
 - Dolby Digital Plus 5.1 or 7.1 bitstream
 - Dolby Digital 5.1 bitstream
- Job editor window controls and options for Dolby TrueHD encoding from a Dolby Atmos master file set, including:
 - **Include Dolby Atmos** button to enable loading a Dolby Atmos master file set.
 - Browser button to load the Dolby Atmos master file set for encoding.
 - **Details** button, to display information about the Dolby Atmos master file set: presentation type, timecode information, encryption, Dialnorm, PCM file, metadata file, bed channel list, total bed channels, and total objects.
 - **Downmix from Dolby Atmos** option (in the **8-Channel Presentation** drop-down menu), which is typically used when encoding from a Dolby Atmos master file set to the 7.1 presentation of a Dolby TrueHD bitstream with Dolby Atmos.
 - **Dolby Atmos render to 7.1 (Lrs, Rrs)** option (in the **8-Channel Presentation** drop-down menu) to encode from a Dolby Atmos master file set to a 7.1 Dolby TrueHD bitstream without Dolby Atmos.

- Job editor window options for Dolby Digital Plus encoding from a Dolby Atmos master file set, including:
 - **Dolby Atmos DD+ Bitstream** option (in the **Channel Config** drop-down menu) to encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos. This option enables **7.1 to 5.1 Downmix** options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core.
 - **Dolby Atmos render to 7.1 (Lrs, Rrs)** and **Dolby Atmos render to 5.1 (Lrs, Rrs)** options (in the **Channel Config** drop-down menu) to encode from a Dolby Atmos master file set to a 7.1 or 5.1 Dolby Digital Plus bitstream without Dolby Atmos.
 - Browser button to load the Dolby Atmos master file set for encoding.
 - **Audio Core Channel Configuration** drop-down menu, to set the channel configuration of the backward-compatible presentation.
- Job editor window options for Dolby Digital encoding from a Dolby Atmos master file set:
 - **Dolby Atmos render to 5.1 (Lrs, Rrs)** option (in the **Channel Config** drop-down menu) to encode from a Dolby Atmos master file set to a 5.1 Dolby Digital bitstream without Dolby Atmos. This option enables **7.1 to 5.1 Downmix** options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core.
- Removal of the **New** button in the **Job Options** section of the Dolby Media Encoder window (the initial Dolby Media Encoder window). For new jobs, create and save a project and then create a job, or open an existing project and add a new job to that project.
- Tool tips for **Destination Path** names and source file path names that exceed the length of the field, so that you can hover over the field and see the full path name.
- Removal of underused channel configuration options from the **Channel Config** list for Dolby Digital and Dolby Digital Plus encoding, to improve user interface (UI) clarity.

The following configuration options have been removed:

- 3.0: L, R, S
- 3.1: L, R, S, LFE
- 3.1: L, C, R, LFE
- 5.1: L, R, C, LFE Ls, Rs (Cs)
- 6.1: L, R, C, LFE, Ls, Rs, Cs
- 6.1: L, R, C, LFE, Ls, Rs, Cvh
- 6.1: L, R, C, LFE, Ls, Rs, Ts
- 7.1: L, R, C, LFE, Ls, Rs, Cvh, Ts
- 7.1: L, R, C, LFE, Ls, Rs, Ts, Cs
- 7.1: L, R, C, LFE, Ls, Rs Cvh Cs
- **Encoder Settings** subpage of the **Encoder Setup**:
 - New data rate options for encoding from a Dolby Atmos master file set to a Dolby Digital Plus bitstream for online media and using 7.1 multichannel configurations that include Lrs and Rrs.
 - **Film Light** is now the default for **Late Night Listening Mode Profile** options for Dolby Digital, Dolby Digital Plus, and Dolby TrueHD. This setting provides moderate compression parameters in the bitstream that the listener may choose to apply during playback.

- **Preprocessing** subpage of the **Encoder Setup**:
 - **Number of Elements** setting: Sets the number of clustered signals plus bed signals of Dolby Atmos material to be produced from spatial coding when encoding from a Dolby master file set to Dolby TrueHD or Dolby Digital Plus.
 - **Bed Configuration** option: To set the bed configuration for spatial coding. Choices include:
 - **LFE (Default)**
 - **LFE,C**
 - **LFE,L,C,R**
 - **LFE,L,C,R,Ls,Rs**
 - **LFE,Ls,Rs,Lrs,Rrs**
 - **LFE,Ls,Rs**
 - **Optimize Data Rate** option: To enable the treatment of intermediate signals in the signal path with additional quantization when encoding from a Dolby Atmos master file set to Dolby TrueHD. This can reduce the peak and average data rates by 10-15% for typical content.
 - Updated version of the Dolby spatial coding algorithm, providing increased dimensional stability and handling of object parameters. This includes the use of a **magnetic centroid** mode, which helps ensure consistent behavior for demanding content when encoding from a Dolby master file set to Dolby TrueHD, Dolby Digital Plus, or Dolby Digital.
- **Timecode** subpage of the **Encoder Setup**
See [Improvements to the Timecode Subpage User Interface](#) for a full list of changes.
- Removal of the **Preview Player** and supporting **Speaker Configuration** menu from Dolby Media Encoder. To listen to encoded material, use Dolby Media Decoder. (The capability of listening to source .atmos master files will be added to Dolby Media Decoder in a future release.)

3.2 Improvements to the Timecode Subpage User Interface

The **Timecode** subpage of the **Encoder Setup** has been updated with new displays, fields, and options to improve working with timecode.

When encoding from a Dolby Atmos master file set with embedded timecode, the UI includes an input timecode display (displaying as **Dolby Atmos Master Offset Time**). When embedded timecode is present, the UI includes the ability to select whether a specified encode start or end time position is relative to the **Embedded Timecode** or **File Position** using the start of the file as 0:00:00:00.



Figure 3-1 Timecode Subpage

You can still use this subpage to enable or disable inclusion of SMPTE time stamps in the output bitstream, as well as to choose the timecode format and starting time for the embedded timecode. You can also specify a time range, which allows you to encode just a portion of the input file, and you can choose to prepend silence to the beginning of the file in increments of frames or milliseconds. Note that prepend silence has been updated to always keep the specified starting timecode displayed in the GUI, at the beginning of the inserted silent period.



Note: When you change the encode range frame rate selection for **Encode Start** and **Encode End** (via the **Encode Start** time range drop-down menu), the start and end times will be reset to the whole frame (or millisecond) value closest to the start and end of the input file. Set the frame rate for the encoding job before entering different start and end points.

Use the <Tab> key to navigate through available entry fields. Modified values will be saved when moving the cursor to the next field using <Tab> or clicking outside the field with the mouse. Press <Enter> to update the entry in a field (and stay in the current field).

The new, updated **Timecode** subpage provides the following:

- **Dolby Atmos Master Offset Time** display of input timecode, which appears when encoding from a Dolby Atmos master file set with embedded timecode. The display shows the initial timecode stamp at the start of an .atmos master (in hrs:min:sec:msec).
- **Specify Time Range to Encode:** Click (check) to specify a range that is a portion of the input file(s) to encode. By default, no range is specified and the entire file is encoded.

- **Time Range Relative To** drop-down menu: This option lets you select whether the specified position is relative to the **Embedded Timecode** position or **File Position** with the start of the file treated as 0:00:00:00. The default is **Embedded Timecode**. If this option is used for files without embedded timecode, a starting timecode of 0:00:00:00 will be used.
- **Encode Start** field: Use this setting to define a valid time point from which to start encoding.
- **Encode End** field: Use this setting to define a valid time point at which to end encoding.
- Encode time range drop-down menu for **Encode Start** and **Encode End** timecode: Use this setting to define the frame rate or real-time value of the encoding. The default is 23.976 fps. There are eight options.

Use one of the ms (millisecond) options if you need to select a start point that is between two video frame boundaries, as follows:

- **Real Time (ms)**: Use this option if you need to edit the encode range in exact milliseconds and the Dolby Atmos file set was recorded in a “real time” SMPTE frame rate (such as 24, 25, or 30 fps). This option treats one second of the entered time as exactly one second of real time.
- **NTSC –0.1%**: Use this option if you need to edit the encode range in “NTSC” milliseconds and the Dolby Atmos master file set was recorded at a slower NTSC frame rate (such as 23.976 or 29.97 fps). This option treats the entered time as slower than real time, by a factor of 1000/1001 of real time, or –0.1%. NTSC seconds last longer than real seconds, so in samples, one second of NTSC time is 48048 samples in duration, not 48000.

Refer to SMPTE ST 12-1:2014, *Time and Control Code* for further information on NTSC versus real time frame rates.

- **Specify New Output Timecode** drop-down menu: Use this menu if you wish to define and include timecode in the output file. Options include **Do not include Timecode**, **Use input Timecode**, and **Specify Timecode**. By default, timecode is not embedded in the encoded output file.
- **Output Timecode** display: The display is grayed out or shows the current output start timecode, dependent on the **Specify New Output Timecode** setting. If **Do not include Timecode** is selected, the field is grayed out. If **Use input Timecode** is selected, this field displays the **Encode Start** timecode from the **Specify Time Range to Encode** section. If **Specify Timecode** is selected, you can change the timecode to any positive timecode value. The default is 00:00:00:00.
- **fps (frame rate)** drop-down menu for **Output Timecode**. This selector is grayed out, or shows the current output frame rate, dependent on the **Specify New Output Timecode** setting. If **Do not include Timecode** is selected, the field is grayed out. If **Use input Timecode** is selected, the field uses the **Specify Time Range to Encode** fps setting, unless it is set to milliseconds, in which case the **Output Timecode** selector can be changed to any frame rate. If **Specify Timecode** is selected, you can also change the frame rate from the default of 23.976.
- **Prepend Silence** field and up/down arrows: Use the up/down arrows to set the duration of silence to add at the beginning of the encoded file. This field accepts valid integers for frames or milliseconds, based on the selected rate (in the **Prepend Silence** drop-down menu). The maximum setting is equivalent to 60 seconds.
- **Prepend Silence** drop-down menu: Use this menu to set whether the **Prepend Silence** value is based on a frame rate (23.976, 24, 25, 29.97, 29.97 df, or 30), **Real Time (ms)**, or **–0.1% NTSC (ms)**.

3.3 New Data Rate Options for Encoding From a Dolby Atmos Master File Set to a Dolby Digital Plus Bitstream for Online Media

Prior to encoding from a Dolby Atmos master file set, to create a Dolby Digital Plus bitstream for online media, you can set the encoding data rate on the **Encoder Settings** subpage. With Dolby Media Encoder v2.0, there are additional data rate options available for 7.1 multichannel configurations that include Lrs and Rrs: 384 kbps, 448 kbps, 504 kbps, 576 kbps, 640 kbps, 704 kbps, 768 kbps, 832 kbps, 896 kbps, 960 kbps, 1008 kbps, 1024 kbps, 1280 kbps, 1408 kbps, 1512 kbps, 1536 kbps, 1664 kbps, 1792 kbps, 1920 kbps, 2048 kbps, 2176 kbps, 2304 kbps, 2432 kbps, 2560 kbps, 2688 kbps, 2816 kbps, 2944 kbps, and 3024 kbps.

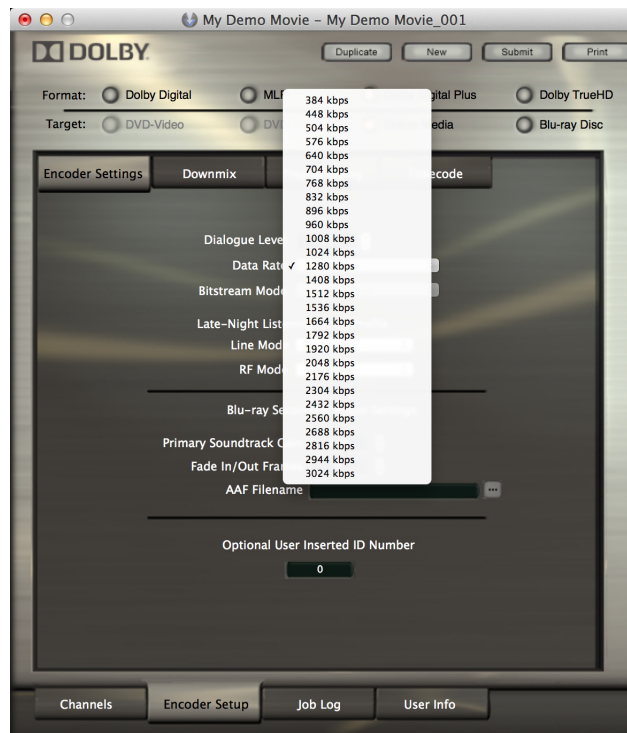


Figure 3-2 Data Rate Options on the Encoder Settings Subpage



Note: Data rates below 640 kbps are not supported when encoding from an .atmos master file set to Dolby Digital Plus with a 7.1 core channel configuration that includes Lvh and Rvh (left and right “vertical height”).

Options are based on the channel configuration of the backward-compatible presentation (as set with the **Audio Core Channel Configuration** drop-down menu).

3.4 Ensuring that Encoded Audio Aligns with Video

Prior to encoding, you may specify the time range for encoding, to tell the encoder to encode just a specific portion of the input file. This is done to ensure that the encoded audio aligns with the encoded video.

When mastering for Dolby Atmos, we recommend adding audio preroll (greater than four seconds) to the audio in Pro Tools®, prior to recording the master file. Then, after importing the .atmos master into Dolby Media Encoder, you set the audio encoding start point to align with the video start frame (the first video frame for encoding, sometimes called the “first

frame of action” or the “first frame of picture”). You can set the encode range relative to the source start timecode that is embedded in the input, or as a “file position” that you specify (with the start of the file treated as time 0:00:00.000).

3.4.1 Specifying the Time Range to Encode, Relative to Embedded Timecode

Prior to encoding, you may specify the time range for encoding, relative to the source start timecode that is embedded in the input.



Figure 3-3 Specifying the Time Range to Encode, Related to Embedded Timecode

To specify the time range to encode, relative to embedded timecode:

1. Click the **Encoder Setup** tab, and then the **Timecode** tab to open the **Timecode** subpage.
2. Click (highlight) the **Specify Time Range to Encode** button.
3. In the **Time Range Relative** to drop-down menu, select **Embedded Timecode**.
If this option is used for files without embedded timecode, a starting timecode of 00:00:00:00 will be used.
4. In the **Encode Start** time range drop-down menu, select the desired fps or millisecond (**ms**) option.
For information on millisecond options, see [Section 3.2](#).
5. For Dolby Atmos material, note the **Dolby Atmos Master Offset Time** time displayed at the top of the window.
This is the start time of the input content, which you will use in the next step.
6. In the **Encode Start** field, enter the timecode at which you wish to begin encoding.
Usually, this time is the same as the desired start of encoded video (first frame of action), for example, 0:59:58:00. This time must be the same or later than the time noted in the previous step.

7. Optional: Enter a different **Encode End** time, which is before the source end time. The **Encode End** field is automatically set to the very end of the input. Change the time only if you wish to trim a portion from the end of your input file.
8. Optional: On the **Timecode** subpage, specify the output file timecode frame rate and start time, as well as prepend silence to the encoded file.

3.4.2 Specifying the Time Range to Encode, Relative to File Position

Prior to encoding, you may specify the time range for encoding as a specified “file position.”



Figure 3-4 Specifying the Time Range to Encode, Related to File Position

To specify time range to encode, relative to file position:

1. Click the **Encoder Setup** tab, and then the **Timecode** tab to open the **Timecode** subpage.
2. Click (highlight) the **Specify Time Range to Encode** button.
3. In the **Time Range Relative** to drop-down menu, select **File Position**.
4. In the **Encode Start** time range drop-down menu, select the desired fps or millisecond (**ms**) option.

For information on millisecond options, see [Section 3.2](#).

5. In the **Encode Start** field, enter the timecode at which you wish to begin encoding. Usually this time should be the same as the desired start of the video (often called the first frame of action). This start point is interpreted relative to the start of the input file, which is treated as time “zero” (ignoring any time stamps in the input) when using **File Position**.

- Optional: Enter a different **Encode End** time, which is before the source end time. The **Encode End** field is automatically set to the very end of the input.

Change the time only if you wish to trim a portion from the end of your input file.

- Optional: On the **Timecode** subpage, specify the output file timecode frame rate and start time, as well as prepend silence to the encoded file.

3.5 Creating a Dolby TrueHD Bitstream with or without Dolby Atmos

You can encode from a Dolby Atmos master file set (home theater master), to create a Dolby TrueHD bitstream with or without Dolby Atmos metadata.

When encoding from a home theater master to Dolby TrueHD, and including one or more independent channel-based presentations from .wav files in the same job, the **Encode Start** and **Encode End** times on the **Timecode** subpage define the encoded duration as well as the start- and end-time points of the time range within the Dolby Atmos master source that will be encoded. However, for the independent presentations from .wav inputs, the **Encode Start** and **Encode End** times define the duration only. The encoding will start from the beginning of the original .wav files. To ensure the encoding captures the desired .wav audio data, trim the beginning of the .wav files in a digital audio workstation prior to importing them into Dolby Media Encoder.



Note: When encoding legacy channel-based presentations only, without including Dolby Atmos, the **Encode Start** and **Encode End** times on the **Timecode** subpage will define both the duration and the sample-accurate start time of the .wav files.

To create a Dolby TrueHD bitstream with Dolby Atmos:

- Launch Dolby Media Encoder.
- Create and save a project and then create a job, or open an existing project and add a new job to that project.
- In the **Format** section, select **Dolby TrueHD**.

In the **Target** section, the system automatically selects **Online Media** and **Blu-ray Disc™**. The system will produce a single encoded file that is compatible with both selected targets. Attributes of these multiple-target files are restricted to the subset of format constraints common to both targets.

4. Select the source material for encoding:
 - a. Click (select) **Include Dolby Atmos** to encode from a Dolby Atmos master file set.



Figure 3-5 Job Editor Window Set Up to Encode to Dolby TrueHD with Dolby Atmos

- b. Click on the **Presentation** configuration drop-down menu for eight-, six-, or two-channel independent presentations, and select the desired channel format.

You may encode independent presentations from unique PCM source files for one, two, or three presentations, provided the program material complexity does not cause the data rate or file size to exceed your format limitations. See “Encoding Configurations” in the *Dolby Media Encoder User’s Manual*.

To encode from a Dolby Atmos master file set to a Dolby TrueHD bitstream with Dolby Atmos, click on the **8-Channel Presentation** drop-down menu and select **Downmix from Dolby Atmos**. To encode from a Dolby Atmos master file set to a channel-based 7.1 Dolby TrueHD bitstream (without Dolby Atmos), select **Dolby Atmos render to 7.1 (Lrs, Rrs)**.

- c. For the remaining dependent presentations, click on the **Presentation** configuration drop-down menu for the desired presentation, and then select the appropriate downmix choice. For example, if independent PCM signals are provided for the eight-channel presentation, a dependent six-channel presentation will be created by downmixing the eight-channel independent presentation.



Note: The **Copy of** options for creating dependent presentations are not supported when encoding from a Dolby Atmos master file set to Dolby TrueHD. **Copy of** options are still supported for creating dependent presentations for encodes to Dolby TrueHD that do not include Dolby Atmos material.

5. Load the master source files by doing the following:
 - a. Click on the browse button to the right of the **Include Dolby Atmos** field.
 - b. In the browser, locate the .atmos master file, highlight it, and click **Open** to add it to the presentation.

This step effectively adds all three master source files to the encoder.



Figure 3-6 Dolby Atmos Master File Set Loaded for Encoding to Dolby TrueHD with Dolby Atmos

6. Add other PCM source files, as needed, for any additional channel-based presentations.



Note: For eight-, six-, or two-channel independent presentations, all input channels must be loaded. Otherwise, the encoder will prompt you to include at least one file for each presentation.



Note: When encoding with both an .atmos master and PCM source signals to include multiple independent presentations in one Dolby TrueHD stream, the timecode settings will apply only to the .atmos master source. Make sure the PCM source signals are already trimmed to the desired start and end time points before loading them.

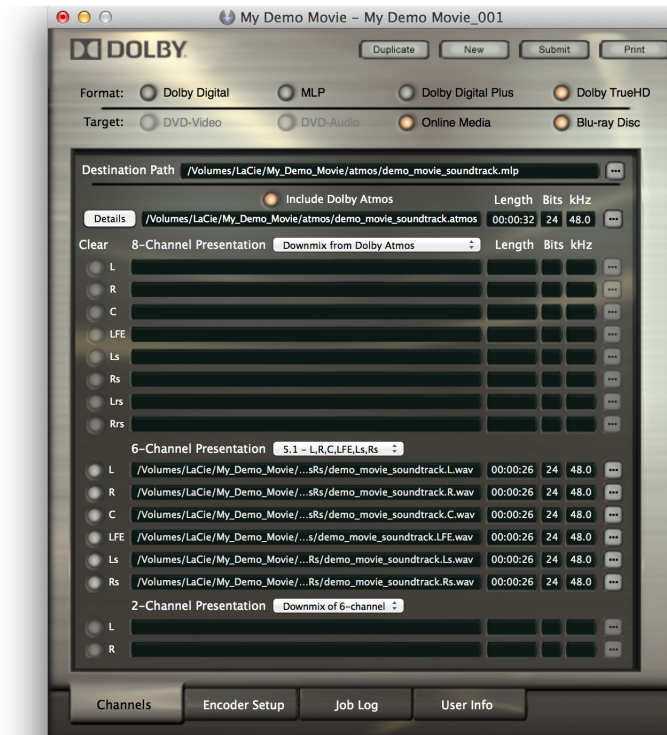


Figure 3-7 Example of Dolby Atmos and PCM Source Files Loaded for Encoding to Dolby TrueHD

7. Verify the destination path in the **Destination Path** field.

After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

8. Set up the Encoder.

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. For more information, refer to the *Dolby Media Encoder User's Manual*.

There are no new settings and parameters on the **Encoder Setup** subpage for **Downmix**.

The **Encoder Settings** subpage includes two settings for a Dolby Atmos Presentation: **Dialogue Level** and **Late Night Listening Mode Profile**. **Film Light** is the default option. This setting provides moderate compression parameters in the bitstream that the listener may choose to apply during playback.

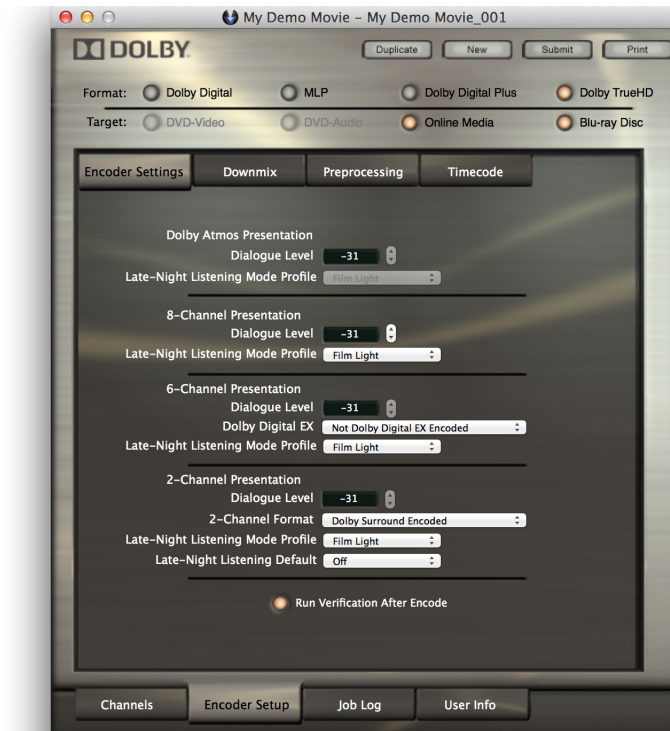


Figure 3-8 Encoder Settings Subpage for Encoding to Dolby TrueHD

The **Preprocessing** subpage includes a **Number of Elements** setting, which represents the number of clustered signals of Dolby Atmos material to be output by spatial coding. Set the **Number of Elements**, based on the following:

- **16:** Use this setting if you have no data space considerations and want the absolute best quality.
- **14:** Use this setting if space allows it and an extra measure of quality is desired.
- **12:** Use this setting for Blu-ray Discs with standard space considerations and excellent sound quality. **12** is the default number of elements, because it provides high quality at an efficient average and peak data rate.

The **Preprocessing** subpage includes three additional spatial coding options: **Bed Configuration**, **Optimize Data Rate**, **Apply Magnetic Centroid**.

Leave the **Bed Configuration** and **Apply Magnetic Centroid** options at their default setting, unless Dolby support recommends different settings.

Use the **Optimize Data Rate** option to apply more aggressive quantization to intermediate signals used in the encoder signal path when encoding from a Dolby Atmos master file set to Dolby TrueHD.



Figure 3-9 Preprocessing Subpage for Encoding to Dolby TrueHD

The **Timecode** subpage has been updated with new displays, fields, and options to improve working with timecode. See [Section 3.2](#).

9. Enter user info for the job.

This is a set of three fields for notes or comments about the job. For more information, refer to the *Dolby Media Encoder User's Manual*.

10. Submit the job.

When prompted, select whether to automatically create a Dolby Digital bitstream and whether to bundle the Dolby TrueHD bitstream, the Dolby Digital bitstream and the Dolby TrueHD encoding log file into a .zip archive after the encoding is done.

For more information, refer to the *Dolby Media Encoder User's Manual*.

When encoding finishes, the status changes to **Completed-Locked**. An .mlp file is now available at the designated destination.

3.6 Creating a Dolby Digital Plus Bitstream with or Without Dolby Atmos

You can encode from a Dolby Atmos Master file set (home theater master), to create a Dolby Digital Plus bitstream with or without Dolby Atmos audio and associated metadata.

Dolby Media Encoder still supports encoding from legacy (.wav) channel-based material to create a Dolby Digital Plus bitstream

To create a Dolby Digital Plus bitstream with Dolby Atmos:

1. Launch Dolby Media Encoder.
2. Create and save a project and then create a job, or open an existing project and add a new job to that project.
3. Select a format and target:
 - a. In the **Format** section, select **Dolby Digital Plus**.
 - b. In the **Target** section, select **Online Media** or **Blu-ray Disc**.
4. Select a channel configuration:
 - a. Click on the **Channel Config** drop-down menu, and select the desired channel format.

To encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos, select **Downmix Atmos DD+ Bitstream**. To encode from a Dolby Atmos master file set to a 7.1 or 5.1 channel Dolby Digital Plus bitstream (without Dolby Atmos), select **Dolby Atmos render to 7.1 (Lrs, Rrs)** or **Dolby Atmos render to 5.1 (Lrs, Rrs)**.

See the section on encoding configurations in the *Dolby Media Encoder User's Manual*.

- b. Set the channel configuration of the backward-compatible presentation by clicking the **Audio Core Channel Configuration** drop-down menu and selecting one of the available configurations.

This menu is not available if you selected **Dolby Atmos render to 7.1 (Lrs, Rrs)** in the previous step (for example, to encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream without Dolby Atmos).

5. Load the master source files by doing the following:
 - a. Click on the browse button to the right of the channel field.
 - b. In the browser, locate the .atmos master file, highlight it, and click **Open** to add it to the presentation.

This step effectively adds all three master source files to the encoder.

6. Verify the destination path in the **Destination Path** field.

After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

7. In the **7.1 to 5.1 Downmix** section, select the type of downmix to be used for the 5.1 backward-compatible core of the Dolby Digital Plus bitstream.

The 5.1 core is made by first rendering the Dolby Atmos audio to 7.1 and then downmixing the four surround channels of the 7.1 into just two surround channels of the 5.1. This is the case whether the core channel configuration is 5.1 or 7.1. The options are **5.1 Dolby Standard** and **5.1 Dolby PLIIx**.

The coefficients for a Dolby Pro Logic® PLIIx downmix from 7.1 to 5.1 are:

$$\begin{aligned}Ls &= Lss + (-1.2 \text{ dB} \times Lrs) + (-6.2 \text{ dB} \times Rrs) \\Rs &= Rss + (-6.2 \text{ dB} \times Lrs) + (-1.2 \text{ dB} \times Rrs)\end{aligned}$$

The coefficients for a Standard downmix from 7.1 to 5.1 are:

$$\begin{aligned}Ls &= 0 \text{ dB} \times Lss + 0 \text{ dB} \times Lrs \\Rs &= 0 \text{ dB} \times Rss + 0 \text{ dB} \times Rrs\end{aligned}$$

8. Set up the encoder.

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. For more information, refer to the *Dolby Media Encoder User's Manual*.

There are no new settings and parameters on the **Encoder Setup** subpage for **Downmix**.

The **Encoder Settings** subpage includes new data rates for online media.

The **Preprocessing** subpage includes a **Number of Elements** setting, which represents the number of clustered signals of Dolby Atmos material to be output by spatial coding. Set the **Number of Elements** to 12 for all Dolby Digital Plus jobs.

- **12** is the default number of elements, and provides the optimal balance between object metadata and perceptual coding data.

The **Preprocessing** subpage includes two additional spatial coding options: **Bed Configuration** and **Apply Magnetic Centroid**. Leave these options at their default setting, unless Dolby support recommends different settings.

The **Timecode** subpage has been updated with new displays, fields, and options to improve working with timecode. See [Section 3.2](#).

9. Enter user info for the job.

For more information, refer to the *Dolby Media Encoder User's Manual*.

10. Submit the job.

For more information, refer to the *Dolby Media Encoder User's Manual*.

When encoding finishes, the status changes to **Completed-Locked**. An .ec3 file (for online media) or .eb3 file (for Blu-ray Disc) is now available at the designated destination.

3.7 Creating a Dolby Digital Bitstream from a Dolby Atmos Master File Set

You can encode audio from a 5.1 channel rendering of a Dolby Atmos Master file set (home theater master), to create a Dolby Digital bitstream.

When encoding from a Dolby Atmos master file set, the bitstream will not carry Dolby Atmos metadata. However, it will sound the same as the 5.1 channel presentation heard when playing a Dolby Digital Plus with Dolby Atmos metadata or Dolby TrueHD bitstream with Dolby Atmos metadata. This bitstream is typically generated for use on Blu-ray Discs, which require a backward-compatible Dolby Digital track to accompany Dolby TrueHD.

Dolby Media Encoder still supports encoding from legacy (.wav) channel-based material to create a Dolby Digital Plus bitstream.

To create a Dolby Digital bitstream from a Dolby Atmos Master file set:

1. Launch Dolby Media Encoder.
2. Create and save a project and then create a job, or open an existing project and add a new job to that project.
3. Select a format and target:
 - a. In the **Format** section, select **Dolby Digital**.
 - b. In the **Target** section, select **DVD-Video**, **Blu-ray Disc**, or both.
4. Select a channel configuration:
 - Click on the **Channel Config** drop-down menu and select the desired channel format: **Dolby Atmos render to 5.1**. See the encoding configurations section in the *Dolby Media Encoder User's Manual*.
5. Load the master source files by doing the following:
 - a. Click on the browse button to the right of the channel field.
 - b. In the browser, locate the .atmos master file, highlight it, and click **Open** to add it to the presentation.

This step effectively adds all three master source files to the encoder.

6. Verify the destination path in the **Destination Path** field.
After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

7. In the **7.1 to 5.1 Downmix** section, select the type of downmix to be used for the 5.1 backward-compatible core of the Dolby Digital bitstream.

The 5.1 core is made by first rendering the Dolby Atmos audio to 7.1 and then downmixing the four surround channels of the 7.1 into just two surround channels of the 5.1. This is the case whether the core channel configuration is 5.1 or 7.1. The options are **5.1 Dolby Standard** and **5.1 Dolby PLIIx**.

The coefficients for a Dolby Pro Logic IIx downmix from 7.1 to 5.1 are:

$$\begin{aligned} Ls &= Lss + (-1.2 \text{ dB} \times Lrs) + (-6.2 \text{ dB} \times Rrs) \\ Rs &= Rss + (-6.2 \text{ dB} \times Lrs) + (-1.2 \text{ dB} \times Rrs) \end{aligned}$$

The coefficients for a Standard downmix from 7.1 to 5.1 are:

$$Ls = 0 \text{ dB} \times Lss + 0 \text{ dB} \times Lrs$$

$$Rs = 0 \text{ dB} \times Rss + 0 \text{ dB} \times Rrs$$

8. Set up the encoder.

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. For more information, refer to the *Dolby Media Encoder User's Manual*.

There are no new settings and parameters on the **Encoder Setup** subpages for **Encoder Settings**, or **Downmix**.

The **Preprocessing** subpage includes two spatial coding options: **Bed Configuration** and **Apply Magnetic Centroid**. Leave these options at their default setting, unless Dolby support recommends different settings.

The **Timecode** subpage has been updated with new displays, fields, and options to improve working with timecode. See [Section 3.2](#).

9. Enter user info for the job.

For more information, refer to the *Dolby Media Encoder User's Manual*.

10. Submit the job.

For more information, refer to the *Dolby Media Encoder User's Manual*.

When encoding finishes, the status changes to **Completed-Locked**. An .ac3 file is now available at the designated destination.

3.8 Unsupported Dolby Media Encoder Features and Functionality

Some legacy encoder features and functionality are unsupported in this release.

The following Dolby Media Encoder features and functionality are not supported with v2.0:

- Encoding from PCM source audio for mono presentations to Dolby TrueHD bitstreams.

To encode from PCM source audio for mono presentations to Dolby TrueHD bitstreams, use the Encoder software installed with Suite v1.1.2 or earlier on a computer that does not have Dolby Media Encoder v2.0 or v1.8.x BETA installed on it. [DMP-3085]

- Large-format web page controls, which can be used to manage jobs on the server over a LAN. The page can be used to view the encoder queue, but not to control jobs.
- **Apply Advanced Upsampling** button for Dolby Atmos encodes (on the **Preprocessing** subpage of the **Encoder Setup**).
- **Punch List** (via the **Punch List** page of the **Job Editor**) has been removed.
- **Preview Player** and supporting **Speaker Configuration** menu have been removed. To listen to encoded material, use Dolby Media Decoder.
- **Pre-copy Source Files to Server** button in the **Submit Jobs** dialog has been removed.

3.9 Dolby Atmos Master Files for Home Theater

Dolby Atmos content for home theater is mastered on a Dolby Rendering and Mastering Unit (RMU) qualified for home theater. Content is stored as a Dolby Atmos Master file set (home theater master). For more information, see the *Dolby Rendering and Mastering Unit Manual*.

Content professionals can then use the Dolby Media Encoder application to encode from the home theater master, to create a Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstream for online streaming or authoring a Blu-ray Disc.



Note: For more information, see the *Dolby Rendering and Mastering Unit Manual*.

3.9.1 Files and System Organization for a Dolby Atmos Home Theater Master

The Dolby Atmos home theater master consists of three files:

- One interleaved PCM file in Core Audio format, with its name structured as <filename>.atmos.audio. This file contains the audio for all bed signals and objects.
- One metadata file with its name structured as <filename>.atmos.metadata. This file contains all the 3D positional coordinates for static and dynamic signals in the .audio file.
- One top-level file with its name structured as <filename>.atmos. This file provides essential information about the presentation contained in the master file set.



Never modify any of the Dolby Atmos content files (including the .atmos file).

Dolby Media Decoder Features and Improvements

Dolby® Media Producer Suite v2.0 updates the Dolby Media Decoder software application to version 2.0.

4.1 Improvements with Dolby Media Decoder

Dolby Media Decoder v2.0 includes new features and improvements, primarily focused on decoding, playing back, and monitoring a Dolby format bitstream encoded from a Dolby Atmos® master file set (home theater master).

Dolby Media Decoder v2.0 includes the following:

- Ability to decode a Dolby format bitstream encoded from a Dolby Atmos master file set:
 - Dolby TrueHD bitstream with Dolby Atmos
 - Dolby Digital Plus™ with Dolby Atmos
 - Dolby TrueHD bitstream
 - Dolby Digital Plus bitstream
 - Dolby Digital bitstream
- **Decode to Files** updates:
 - All .wav files are saved as 24-bit audio files.
 - 4 GB and larger files are decoded to the RF64 standard.
 - Supports saving presentations up to 7.1 from a Dolby Atmos bitstream.
- Visual feedback of loading activity when loading a file for playback (in the **Building Index for Audio File** window), plus a **Cancel** button to cancel loading.
- Improved playback and syncing of audio and QuickTime® movies to handle cases where audio and video files begin or end at different times. The decoder now starts playback at the earliest file and stops playback at the end of the last file to ensure the full timeline is decoded.
- Visual feedback to indicate the type of media currently being played back:
 - In the transport window (upper-right corner of the timecode counter): **A** for audio only, **V** for video only, or **AV** for audio/video.
 - In the QuickTime window (footer bar): **Audio**, **Video**, or **Audio/Video**.

- Transport window updates:
 - Ability to drag and drop a QuickTime .mov file to the transport window, in order to load the file into the decoder.
 - Timecode slider optimized to be audio-sample-accurate when dragging with a mouse. While dragging, the following occurs:
 - Timecode counter shows the current position.
 - Video updates to the current position.
 - Meter display clears, and there is no sound.
- **Dolby Atmos** indication in the information display, to confirm Dolby Digital Plus and Dolby TrueHD bitstreams carrying Dolby Atmos audio and metadata.

The indication text is green for all bitstreams with Dolby Atmos objects and metadata. The indication is orange if the additional ADDBSI header information in Dolby Digital Plus is not present. The decoder supports playback for both signals.

- **(Downmix)** indication in the information display (third line), to confirm Dolby TrueHD or MLP presentations that are downmixes.
- Data rate indication on the right-side of the information display, below the Dialog level.
- Transport counter controls (time display field) optimized to be audio-sample-accurate when entering a time value.
- Timecode entries in the timecode counter are applied when window focus is changed (such as clicking outside the transport window or launching another window).



Figure 4-1 Transport Window with Updated Indicators

- Meters/gain window updates:
 - 16-channel metering, to support speaker configurations of up to 16 speakers.
 - Metering labels provide improved feedback for all encoded material, as they now show 0, -3, -6, -10, -20, -40 and -60 dB levels. Signal is displayed as follows: in green when below -6 dB; in yellow from -6 dB to -0.000005 dB; in red when at or above 0 dB.

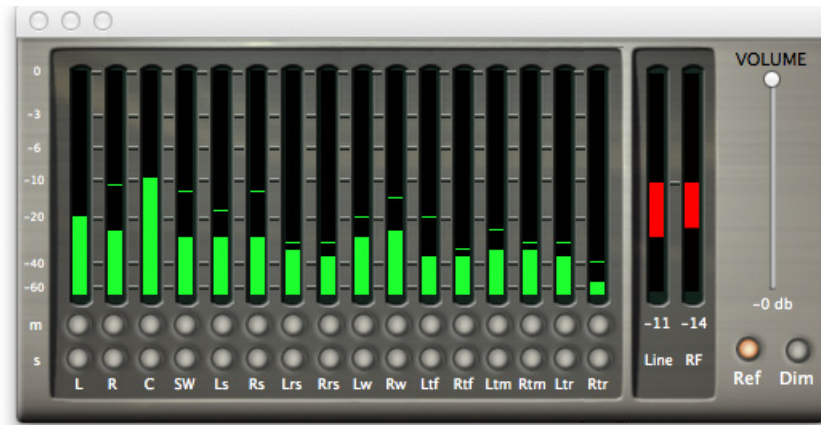


Figure 4-2 Meters/Gain Window with Updated Metering

- Metadata window update:
 - **Bitstream Type** in the **Metadata** field, to note whether the encoded file bitstream is Dolby Atmos or channel-based.
 - Window header display of **Dolby Digital Plus w/ Dolby Atmos** for Dolby Digital Plus with Dolby Atmos encoded files.
 - Display of metadata for Dolby TrueHD with Dolby Atmos or Dolby Digital Plus with Dolby Atmos encoded files, including the number of **Encoded Elements** (the **Number of Elements** set during encoding with the Dolby Media Encoder) and the **Dolby Atmos ADDBSI Flag**.

The **Dolby Atmos ADDBSI Flag** value is **Yes** when the file contains the additional bitstream flag that verifies that the bitstream is a Dolby Atmos bitstream.



Figure 4-3 Metadata Window for Dolby Digital Plus with Dolby Atmos

- QuickTime movie screen window update:
 - Ability to drag and drop a QuickTime .mov file onto the screen, in order to load the file into the decoder
 - Improved movie handling on Retina® displays
 - Improved nonbordered UI, so that the video better fills up the canvas



Figure 4-4 QuickTime Movie Screen Window After Drag and Drop of a .mov

- **Preferences** window updates:
 - Ability to create listening modes with up to 16 speaker channels (for example, to audition Dolby Digital Plus bitstreams carrying Dolby Atmos audio and metadata)

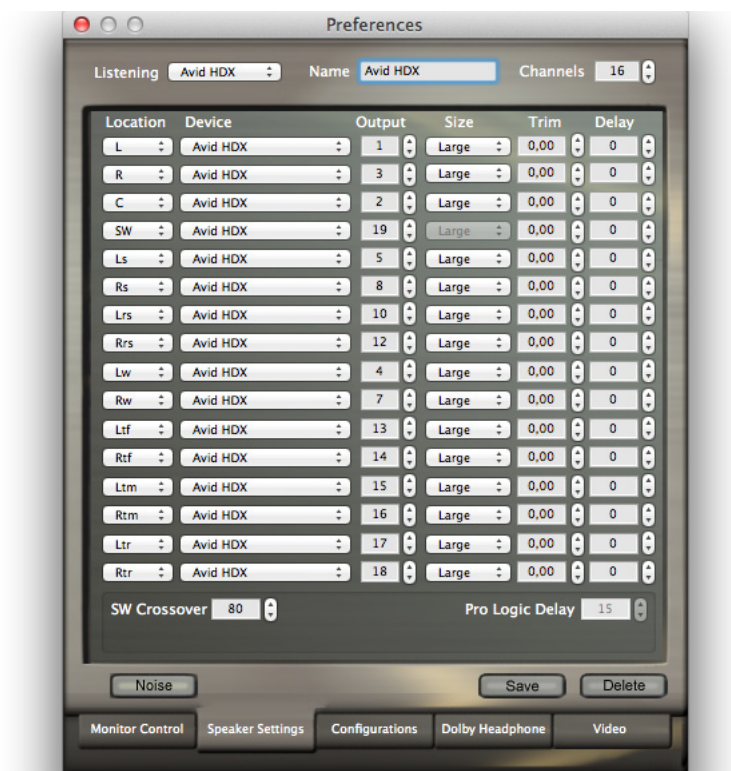


Figure 4-5 Preferences Window and Speaker Settings Page Updates

- Support for new Dolby Atmos overhead speaker locations:

- Ltf/Rtf: left/right top front
- Ltm/Rtm: left/right top mid
- Ltr/Rtr: left/right top rear
- Lrh/Rrh: left/right rear height

Dolby Digital Plus bitstreams carrying Dolby Atmos audio and associated metadata also support Lfh/Rfh (or Lvh/Rvh) left/right front height speaker locations. These locations are also supported by Dolby Digital, Dolby Digital Plus, and Dolby True HD bitstreams.

- Support for new Dolby Atmos ear-level speaker locations:

- Lsc/Rsc: left/right screen channel
- Ls1/Rs1: left/right surrounds 15 degrees forward of standard Ls and Rs locations
- Ls2/Rs1: left/right surrounds 15 degrees rearward of standard Ls and Rs locations
- Lrs1/Rrs1: left/right surrounds 15 degrees forward of standard Lrs and Rrs locations
- Lrs2/Rrs2: left/right surrounds 15 degrees rearward of standard Lrs and Rrs locations
- Lcs/Rcs: left/right surrounds 15 degrees to either side of the standard Cs (center surround) location

- Speaker configuration defaults to stereo with built-in output when first launching the application so that the decoder is ready for immediate use.

- **Window** menu keyboard shortcuts:

- Metadata window opens by pressing <M> on the keyboard.
- QuickTime movie screen windows opens by pressing <V> on the keyboard.

4.2 Unsupported Dolby Media Decoder Features and Functionality

Some legacy decoder features and functionality are currently unsupported in this release. Additionally, the Dolby Media Decoder v2.0 does not have the ability to load a Dolby Atmos master file set (.atmos).

The following Dolby Media Decoder features and functionality are not supported and have been grayed out:

- Upmixing controls (in the monitor window).
- **Preferences** window:
 - **Monitor Control** page: **Pro Logic II** and **Decode Control** settings are grayed out.
 - **Dolby Headphone** page: All controls are grayed out.
 - **Video** page: All controls are grayed out.

The following features have been removed:

- Two Display mode options (from the transport window): **TC Start** and **TC Offset**
- **Mixing Level (Max SPL per channel)** data (from the metadata window).

List of Known Issues

The new software has some known issues that may affect your workflow or data. We recommend reviewing these issues prior to using the software. Workarounds are provided, where applicable.

5.1 Dolby Media Encoder Known Issues

Dolby® Media Encoder has some known issues in this release, some of which have workarounds and solutions.

- Occasionally, the iLok™ inserted in your computer may not be recognized by Dolby Media Encoder, and the application will display “Activation Error” messages repeatedly. If this happens, launch the iLok License Manager application (which is required for iLok use) and verify that the manager sees your iLok. Once the iLok presence is verified in the License Manager, Dolby Media Encoder should start normally. If the License Manager continues to not recognize your iLok, there may be issues with the iLok or the required iLok drivers. [DMP-2075]
- Occasionally after a system reboot on the server host, the client will lose contact with the Dolby Media Encoder Server application. The connection can be checked in the job list window by selecting (highlighting) the **Encoder Queue** radio button, and looking for your desired server host in the **Server** drop-down menu (located to the left of the **Project** and **Encoder Queue** radio buttons). If the client returns an error indicating that the server is not found, reboot the system again. [DMP-2061]
- Currently, the Dolby Media Encoder Server running on a remote server computer will not be accessible if the computer name assigned to that computer is written using UTF-8 or UTF-16 characters. (To see the computer name, choose **System Preferences > Sharing**). To ensure the remote server can be accessed, use ASCII characters for the computer name. [DMP-3331]
- The Dolby Media Encoder Server application is designed to automatically restart if it crashes. However, if it crashes while performing an encoding job, the automatic restart behavior may not occur. If the server fails to restart after a crash occurs, restart the computer on which it is running. [DMP-3321]
- In some rare occasions, it may be necessary to force the Dolby Media Encoder Server application to quit using the Mac Activity Monitor (located in Applications/Utilities). This may result in two instances of the Dolby Media Encoder Server launching. If this happens, force quit the second instance before submitting jobs for encoding. [DME-3360]
- In the **Encoder Servers** dialog, the **IP Address** field for a server does not always update or appear after modifying the **Server** field. As a workaround, double-click the new server IP address, or right-click or <Control>-click the server in the **Server** field and select the desired new server location from the list shown. Entries on this list are discovered with the Apple Bonjour service and should be recognized by your client without any further steps. [DMP-3363, DMP-3322]

- The large-format web page, which can be used to manage jobs on the server over a LAN, is not supported with Dolby Media Encoder v2.0. The large-format web page interface will be enabled in a future release. [DMP-2132, DMP-2377]
- Dolby Media Encoder works best with only one active project at a time. Occasionally, after creating and closing multiple projects in the project window, the encoder will be unable to create a new project. If this occurs, close Dolby Media Encoder, re-launch it, and then open the desired project. [DMP-2161]
- Occasionally, the Dolby Media Encoder job list window project selection mechanism will begin to show closed projects, no project name, or projects other than the one selected in the **Filename** drop-down menu. If this occurs, save and close any open job editor windows, quit Dolby Media Encoder, and then re-launch it. [DMP-3348], [DMP-3126]
- The **Pause** option in the Encoder Queue does not keep a submitted Dolby TrueHD job in the paused state if the Dolby Media Encoder Server is restarted or rebooted. The job will start running again and run to completion unless **Pause** is chosen a second time. [DMP-3320]
- In the Encoder Queue, attempting to remove a job that was created in a project that was closed may result in prompts for "Administrator Password." To avoid this, open the project file (.dpp) in which the job was created and then delete the job from the Encoder Queue. [DMP-2593]
- When preparing to encode from a Dolby Atmos master file to a Dolby TrueHD bitstream with multiple independent presentations, the input signals must be of the same total duration. If the input signals are of different durations, the Dolby Media Encoder Client application may incorrectly appear to complete the encode process, no .mlp output will be created, and the log will not indicate a problem. If this happens, adjust the input signals to have the same duration, and set up a new job with the adjusted input signals. [DMP-1984]
- The text field used to display the path and file name for an .atmos master file loaded for encoding does not currently support proper display of UTF-8 or UTF-16 characters. In most cases, the job can still be submitted and encoded successfully. For improved readability, use ASCII standard characters for the .atmos file name and file path. [DMP-3338]
- In the job editor window, the **Destination Path** can be accidentally set incorrectly when the **Destination Path** field is edited manually. For example, the file name may be missing its extension. Instead, browse to the desired location and enter the file name using the OS browser. [DMP-2497]
- In the job editor window, the **Destination Path** does not support the ampersand (&) character for path or file names. [DMP-2073]
- The Dolby Media Encoder Server may remove source files loaded into channel fields if the audio compression format is changed while the job is being created or edited. Be sure to select the desired format before entering file path information for the job. [DMP-2001]
- The Dolby advanced 96 kHz upsampling technology for Dolby TrueHD encoding is not yet available for Dolby Atmos. [DMP-2079]
- If a job template has been created to encode from Dolby Atmos master files to a Dolby Digital Plus™ bitstream with **Channel Config** set to 5.1, the template will reset this parameter to 7.1. Check the **Channel Config** setting to make sure it is correct for your job, even if creating it from a template. [DMP-2178]
- If a Dolby Atmos encode job is submitted with corrupted data in any of the three files of the .atmos master, the encoder reports an error message: `Server: DD/DD+ Atmos Error: pipeline.init`. If this occurs, check the files in the .atmos master file set for integrity. If needed, obtain a new copy of the master. [DMP-3382]

- If you attempt to encode a job without an iLok inserted, the job will fail and the client will report an error message: `no slave`. Most of the time, inserting the iLok will allow a duplicate for that job to be submitted. If the server does not start properly encoding the job after the iLok is replaced, reboot the computer hosting the server application. [DMP-3362], [DMP-3203]
- When encoding Dolby TrueHD audio from 96 kHz or 192 kHz PCM source files and specifying output timecode in a frame rate of 29.97 df (drop-frame), the specified value will drop by one half when the 29.97 df rate is selected or when the value is entered after selecting the frame rate. As a workaround, first select 29.97 df, and then enter a timecode value which is twice the desired timecode. [DMP-3356]
- When encoding a 5.1-channel Dolby Digital Plus signal for use as a secondary audio stream for Blu-ray Disc authoring, note that the **Preferred Stereo Downmix** parameter selection mechanism is not working properly. The flag will be set to **Stereo** in the bitstream even if the user control is set to **Surround**. This means that a decoder playing back this stream when set to **Auto** downmixing will use an Lt/Rt downmix, instead of an Lo/Ro downmix. This can shift the image of the rear channels as presented in a 2-channel downmix. [DMP-3317]
- Currently, there are difficulties running encoding jobs for source or destination file or path names that use UTF-8 characters on a remote server if automounting is used. To avoid this issue, manually mount the source and destination directories to the computers running the Dolby Media Encoder Client and Dolby Media Encoder Server applications, and set up path translation entries as needed. Alternatively, run the server application locally, on the same computer as the client. [DMP-3330]
- When encoding, the required amount of available hard disk drive space for temporary file use can be large, particularly for encoding to Dolby TrueHD bitstreams. We recommend that your system drive (where temporary files are stored) have at least 17 GB of free space per hour of content to be encoded. This typically means greater than or equal to 34 GB of free space for a two-hour film. This does not include the space for the input master or the output bitstream. We recommend that the input and output files reside on a “data” drive, separate from the system drive, whenever possible. [DMP-1986]
- If a job is encoding and access to the source files is lost or disconnected during the encode process, the encode process will be interrupted, and the job log will report messages (such as “Error accessing file: '<path>/<file name>' No such file or directory”). However, the last line of the log will incorrectly report that the job has completed, and the job list in the project window will also say that the job has completed. To avoid this issue, never disconnect disk drives and network file access during encoding. [DMP-2122]
- The encoding progress display at the bottom of the job editor window may stop updating if multiple projects are open at the same time. Encoding progress can be monitored in the job list window, as described in the *Dolby Media Encoder User's Manual*. [DMP-2610]
- If the **Encode Start** and **Encode End** time points for encoding a specific portion of the audio source files define a range that is longer than the duration of the source files, the encoding job will not run. Ensure that the total duration for encoding is the same length or less than the duration of the source audio. [DMP-2609]
- Hot Jobs created with previous versions of Dolby Media Encoder are not compatible with Dolby Media Encoder version 2.0, and will need to be created again. [DMP-671]
- Hot Job encoding does not support encoding using a remote server, or encoding to bitstreams with Dolby Atmos® metadata. For channel-based hot jobs, use the local server. [DMP-3228]

- Attempting to open a Hot Job for editing from the Finder when the Dolby Media Encoder application is not already running has been found to sometimes not launch the encoder. To avoid this, launch the encoder, and then open the Hot Job from the Finder. [DMP-3231]
- Hot Job encoding does not support encoding to a remote server, or encoding to Dolby TrueHD or Dolby Digital Plus with Dolby Atmos bitstreams and metadata. For channel-based hot jobs, use the local server. [DMP-3228]
- The Hot Job automated encoding feature currently does not support encoding of two-channel Dolby TrueHD signals. Encoding with a Hot Job for 5.1 and 7.1 channel Dolby TrueHD configurations is supported. [DMP-3233]
- Currently, when a Hot Job is opened and the **Job Name** field on the **User Info** page is edited, the client will create a new Hot Job on the desktop with this name. The old job will remain unmodified and remain in its original location. If the old job is no longer needed, delete it. [DMP-3311]

5.2 Dolby Media Decoder Known Issues

Dolby Media Decoder has some known issues in this release, some of which have workarounds and solutions.

The Dolby Media Decoder has the following known issues:

- When installing Dolby Media Producer, the **Installer Log** window will show information about other applications that rely on the Apple Installer. This includes installer packages, as well as the Mac® OS® software update feature. To minimize this, run only one installer at a time, and close it as soon as the installation is complete. [DMP-2751]
- After restarting the computer, the Dolby Media Decoder **Open Recent** dialog in the File menu drop-down may clear and no recent files will be shown. [DMP-2835]
- It has been found that some audio interfaces connected via Firewire may not play back audio encoded at a 192 kHz sample rate. If possible, connect the device via USB 2.0 instead, or use alternative audio hardware. [DMP-3263]
- Dolby Media Decoder provides an error message if an Avid® HD device is no longer accessible (disconnected, or powered down) and you press Play, or the decoder can experience instability if the device is not found. [DMP-3002]

As a workaround, do the following:

- a. Close Dolby Media Decoder.

If the Dolby Media Decoder is hung, press <Command>+<Option>+<Escape>, and then select Dolby Media Decoder and click Force Quit.

- b. Verify that the Avid HD device is connected to the system.
- c. Verify that the Avid HD device is on.
- d. Open the CoreAudio Manager and validate that the manager is connected to the device.
- e. Relaunch Dolby Media Decoder.
- f. Load an Avid HD configuration from the **Configurations** page in the **Preferences** window.
- g. Load an encoded file, and press play.
- h. If Dolby Media Decoder does not recognize the Avid HD device, shut down the computer and HD device, and then follow the Avid instructions for relaunching the device and the computer.

- In the Dolby Media Decoder transport window, when you set a specific timecode start or offset (either by editing the value in the timecode counter or by dragging the slider), the video may start to play from a different frame than the one specified. This is because the Decoder only supports setting timecode start or offset to audio frame boundaries, which are not always aligned to video frames. [DMP-2808, DMP-2596]
- In some cases, resizing the video playback window can cause the video to go black. If this occurs, start playback and the video image will appear again. [DMP-3213]
- The **Decode to Files** feature currently works only for encoded files that use standard 5.1 and 7.1 channel layouts. Output from encoded files using other channel layouts may produce results that exhibit a mixing of signals between multiple output .wav files. [DMP-3009]